



Fiona Stanley Fremantle Hospitals Group

Policy and Procedure

Electroconvulsive Therapy – Recovery and Discharge

Reference #: FSFH-MENH-POL-0003

Scope

Site	Service/Department/Unit	Disciplines
Fremantle Hospital	Mental Health Service	Medical, Nursing, Allied Health

1. Introduction

This policy and procedure outlines the care and management of patients recovering from Electroconvulsive Therapy (ECT) and the requirements for criteria led discharge.

2. Terminology

Electroconvulsive Therapy	Application of an electric current to specific areas of the head to produce generalised seizure; modified by administration of a short acting general anaesthetic and a muscle relaxing agent optimally timed to reduce consciousness and modify muscle stimulation.
Montreal Cognitive Assessment (MoCA)	A widely used screening assessment for detecting cognitive impairment. The MoCA is performed pre-ECT, post sixth ECT treatment session and after 12 ECT sessions.
First stage recovery	Occurs in the recovery room and lasts until the patient is alert, orientated, has recovered motor function and protective reflexes.
Second stage recovery	Occurs in the ECT discharge lounge or on the ward and continues until the discharge of day procedure patients or the resumption of regular ward routine for inpatients.
Third stage recovery	Includes full physiological, psychological, and cognitive recovery which may take weeks or months.
Mental state	A systematic method for observing, examining, and recording

examination (MSE)	a person's appearance, behaviour, speech, mood, affect, thought form, thought content, perception, cognition, insight, and risk. The MSE follows a set schema all mental health professionals are trained to use.
Patient centred care	Health care that is respectful of and responsive to the patient's preferences, needs and values.
iSoBAR	Mnemonic used as a tool to guide the context and structure of handover in a manner that suits the clinical context. Identify, Situation, Observations, Background, Agree plan, Read back.

3. Policy

- ECT will be administered in compliance with the Mental Health Act (MHA) 2014, the Chief Psychiatrists Practice Standards for the Administration of ECT 2015 and the Chief Psychiatrists Guidelines for the use of Electroconvulsive Therapy in Western Australia 2006.
- Day procedure patients who meet the discharge criteria may be discharged from the ECT discharge lounge by nursing staff two hours post treatment.
- Day procedure patients who do not meet the discharge criteria must be reviewed by a medical officer.
- A Registered Nurse (RN) competent in Hospital Advanced Life Support must be available in the recovery area during the first stage of recovery.
- Nursing staff competent in Hospital Life Support and trained in ECT recovery must oversee the second stage of recovery.
- The treating or referring Psychiatrist is responsible for documenting approval for ECT Criteria Led Discharge for the patient, managing patient treatment and fulfilling the requirements of the MHA 2014.
- The ECT Consultant Psychiatrist is responsible for matters pertaining to the ECT service.

4. Procedure

4.1. Patient centred care

- To provide care that is patient centred and facilitates the best outcomes for the patient, the ECT service will:
 - Establish formal and informal methods of communication between the ECT service, the treating team, ward staff, the patient and other stakeholders.

- utilise approved interpreter services to assist patients from culturally and linguistically diverse backgrounds
- inform the patient of their rights in respect to ECT
- provide education pre and post ECT procedure in a format that is meaningful to the patient/carer.

4.2. ECT side effects

- ECT performed under general anaesthetic is generally accepted as a safe procedure, with a mortality rate of 0.002%. Inform the patient and carer where appropriate of potential side effects of ECT. Potential side effects may include:
 - mild and transient headache
 - nausea
 - dry mouth
 - myalgia
 - transient drowsiness
 - reversible memory deficits
 - post ictal delirium/confusion/agitation (generally resolves within 5 - 45 minutes)
 - transient fluctuation in pulse rate or blood pressure
 - transient arrhythmias
- Rarely side effects may include:
 - myocardial infarction
 - delayed inter-ictal delirium of prolonged duration
 - enduring memory deficits
- Side effects will be treated symptomatically
- Adverse cognitive effects will be minimised by review of stimulus parameters and electrode placement.
- Adult/older adult patients with pre-existing risk factors will have appropriate investigation and assessment prior to commencing ECT. Refer to the Electroconvulsive Therapy Policy and Procedure.

4.3. First stage recovery

- Support the patient's airway on transfer from the treatment room to the recovery room and commence supplemental oxygen therapy immediately on arrival.
- A Post Anaesthetic Care Unit (PACU) nurse will provide 1:1 care in the recovery room including cardiac monitoring and continuous observation until the

patient has recovered protective reflexes to maintain their own airway and breathing.

- Record observations on arrival in the recovery room and at least 10 minutely for a minimum of 30 minutes. After 30 minutes the frequency and duration of observations will be dictated by the patient's clinical condition. Observations include:
 - pulse
 - oxygen saturation
 - end tidal carbon dioxide
 - blood pressure
 - respiratory rate
- In addition, the following will be documented:
 - temperature. Medical review is required if $<36^{\circ}\text{C}$ or $> 37.5^{\circ}\text{C}$.
 - sedation level
 - conscious state
 - motor strength.

Provide clinical handover using the iSoBAR format at every change of care from one clinician to another.

- Document physiological observations immediately post ECT on the Integrated ECT/Anaesthesia Record:
- The last set of physiological observations will be transcribed onto the Adult Observation and Response Chart (AORC).
- A cognitive assessment will be completed 30 minutes post ECT.
- Manage any nausea and vomiting symptomatically.
- Patients experiencing pain or with any other concerns must be reviewed by a medical officer.
- Prior to transfer to second stage recovery the following criteria must be met:
 - physiological observations are within normal physiological parameters
 - conscious state assessed as alert or at baseline level of consciousness
 - orientation is assessed and documented
 - motor function adequate to lift head from pillow for five seconds on request
 - able to breathe deeply and cough
 - pain and nausea have been relieved to an acceptable level for the patient

- the patient is clean, dry, warm, comfortable, and sitting upright
- intravenous cannula has been removed unless specified
- a wheelchair is available to transfer the patient to the ward
- the patients' property and documentation is available to be returned to the ward.

4.4. Second stage recovery

- The Anaesthetist will remain in the ECT suite until patients are adequately recovered.

A decreased conscious state or confusion may indicate delirium or seizure activity. A delay in the progression from recovery to the ward may be necessary in the presence of post ictal delirium, continued seizure activity, drug reaction, sedation level, reversal agents and/or uncontrolled nausea or pain. Request medical review.

- On return to the ward/discharge lounge:
 - Monitor and document physiological observations and conscious state on the AORC at least every 30 minutes for two hours post procedure or more frequently if clinically indicated.
 - escalate any deviation to the Adult Deterioration Detection System (ADDS) score according to instructions on the AORC.
 - monitor orientation until it is comparable to pre-treatment.
 - reassure and reorient the patient.
 - offer diet and fluids to the patient.
 - if no risk is identified administer regular medications.
 - regularly assess pain and nausea and treat appropriately.

4.5. Discharge criteria

- Prior to resuming routine ward observations for inpatients or discharge for approved day procedure patients all of the following criteria must be met:
 - it is at least two hours since completion of the ECT procedure
 - ADDS score is zero or within parameters set for the patient for at least one hour
 - a MSE has been completed and documented in the integrated progress notes
 - the Danish Society of Anaesthesiology and Intensive Care Medicine (DASAIM)³ Discharge Assessment Tool total score of all criteria is four or less and there is no single score above one and has been documented in the patient's medical record. Refer to [Appendix 1](#)

- the Abbreviated Mental Test Score (AMTS) 4 is greater than or equal to eight and has been documented in the patient's medical record. Refer to [Appendix 2](#)
- the treating Consultant has documented the patient is suitable for criteria led discharged.

Patients who do not meet the DASAIM total score must be reviewed by a medical officer.

Patients who do not have documented approval for Criteria Led Discharge must be reviewed by a medical officer.

Approval for Criteria Led Discharge may be withdrawn at any time.

Medical review prior to discharge may be requested for any relevant concern.

- Day procedure may only be discharged if the following criteria are met:
 - patient has a responsible adult to take them home
 - patient has access to carer/support for the next 12 – 24 hours
 - patient has been advised not to drive or use heavy machinery for 24 hours
 - If any of these criteria are not met the risk will be assessed and arrangements made for admission to Hampton House overnight or the procedure cancelled if clinically indicated.
- Day procedure patients must be provided with an ECT discharge information and advice sheet including documentation of their next treatment and review date if known.

4.6. Treatment evaluation

- The quality of the medically induced seizure will be assessed and documented by the ECT Treating Psychiatrist/Registrar following each treatment and used to guide recommendations including dose.
- A MoCA will be documented on commencing treatment, after six treatments and at the completion of treatment 12 by a medical officer or the ECT coordinator.
- For day procedure ECT a post discharge telephone call may be conducted by the ECT Coordinator or Community Treating Team care coordinator.
- All clinical incidents will be notified in the Datix Clinical Incident Management System (CIMS) according to the WA Health [Clinical Incident Management Policy 2019](#).

5. Compliance/performing monitoring

The ECT service will maintain an ECT Register and will report ECT data to the Office of the Chief Psychiatrist on a monthly basis as required by the Mental Health Act 2014.

Clinical incidents notified in Datix will be regularly monitored and reviewed by the FH ECT working group

6. Related Policy Documents

WA Health:

- [Clinical incident management policy MP 0122/19](#)
- [Chief Psychiatrists Practice Standards for the Administration of ECT 2015](#)
- [The Chief Psychiatrist guidelines for the use of Electroconvulsive Therapy in Western Australia 2006.](#)

FSFHG Policies

- Cognitive screening of the older person (FSFH-HW-POL-0028)
- Clinical handover (FSFH-HW-POL-0035)
- Clinical documentation (FSFH-HW-POL-0039)
- Patient Identification (FSFH-HW-POL-0034)
- Observation and response chart (FSFH-HW-POL-0032)
- Adult Seizure Management (FSFH-HW-GUI-0017)
- Post-operative discharge telephone follow (FSFH-DSU-GUI-000)
- Recognising and responding to clinical deterioration (FSFH-HW-POL-0033)
- Discharge criteria for day stay patients (FSFH-DSU-GUI-0001)
- Electroconvulsive Therapy (FSFH-MENH-POL-0014)
- Electroconvulsive therapy Coordinators role (FSFH-MENH-POL-0013)
- Electroconvulsive Therapy Registrar Role (FSFH-MENH-POL-0015)

7. Related Legislation

[Mental Health Act 2014](#)

8. Related Standards

NSQHS:

- Clinical Governance
- Partnering with consumers

- Communicating for Safety
- Recognising and responding to acute deterioration

The Chief Psychiatrist Standards for Clinical care:

- Care planning

[Royal Australian New Zealand College of Psychiatrists \(RANCP\) Professional Practice Guidelines for the administration of electroconvulsive therapy \(ECT\) 2019](#)

9. Bibliography

1. Greenburge R, Kellner C. Electroconvulsive Therapy: a selected review. Am J Geriatr Psychiatry [internet]. 2005 [cited 2017 May 08]; 13(4):268-281. doi: 10.1097/00019442-200504000-00002
2. Andrade C, Arumugham S, Thirthalli J. Adverse Effects of Electroconvulsive Therapy. Psychiatr Clin Am [internet]. 2016 [cited 2017 May 04]; 39(1):513-530. doi: 10.1016/j.psc.2016.04.004
3. Philip N, Street M, Kent B, Cadeddu M. Post-anaesthetic discharge scoring criteria: key findings from a systematic review. Int J Evid Based Health [internet]. 2013 [cited 2017 May 04]; 11:275-284. doi:10.1111/1744-1609.12044
4. Aged Care Network Sub-group for Delirium Services. Delirium model of care. Perth: Western Australian Department of Health, Aged Care Network: 2008

10. Authorisation

EXECUTIVE SPONSOR: Service 5 Service Director					
Version	Date Issued	Compiled/Revised By	Committee/Consumer Group Consulted	Endorsed By	Revision due
1	09/2018	ECT Coordinator	FSFHG SQR Committee FSFHG NMPGC	FSFHG Policy Committee	08/2021
2	11/2020	ECT Coordinator	FSFHG SQRC Service 5	FSFHG Policy Committee	10/2023
3	11/2023	ECT Coordinator		FSFHG SQRC Service 5	11/2027

11. Appendices

11.1. Appendix 1: Danish Society of Anaesthesiology and Intensive Care Medicine (DASAIM) Discharge assessment tool (2010).

Modality	Score	Criteria
Sedation (nurse evaluation)		
	0	The patient is fully awake
	1	The patient is asleep, aroused by verbal stimuli
	2	The patient is asleep, aroused by physical stimuli
	3	The patient is asleep, cannot be aroused
Respiration rate (nurse count)		
	0	Regular rate > 10
	1	Snoring, 10 > RR > 30
	2	RR < 10 or RR > 30/min
	3	Periods of apnoea or obstructive pattern
Oxygen saturation, no supplementary oxygen for 10 mins		
	0	SPO2 ≥ 94%
	1	90% ≤ SPO2 < 94%
	2	85% ≤ SPO2 < 90%
	3	SPO2 < 85%
Systolic blood pressure (automatic NIBP)		
	0	SBP ≥ 100mmHg
	1	90mmHg ≤ SBP < 100mmHg
	2	80mmHg ≤ SBP < 90mmHg or SBP > 220mmHg
	3	SBP < 80 mmHg
Heart rate (automatically derived from ECG)		

0	$50 < \text{HR} \leq 100$
1	$100 < \text{HR} \leq 120$
2	$40 < \text{HR} \leq 50$ or $120 < \text{HR} \leq 130$
3	$\text{HR} < 40$ or $\text{HR} > 130$

Pain (patient evaluation)

0	$50 < \text{HR} \leq 100$
1	$100 < \text{HR} \leq 120$
2	$40 < \text{HR} \leq 50$ or $120 < \text{HR} \leq 130$
3	$\text{HR} < 40$ or $\text{HR} > 130$

Nausea (patient evaluation and nurse observation)

0	No nausea and not vomiting
1	Light nausea or vomiting without previous nausea
2	Moderate nausea and/or vomiting
3	Severe nausea and/or recurring vomiting

TOTAL SCORE= _____ Document score in the patient's medical record

RR = respiration rate; SPO2 = oxygen saturation; SBP systolic blood pressure; HR = heart rate; NIBP = non-invasive blood pressure; ECG = electrocardiography

11.2. Appendix 2: Abbreviated Mental Test Score (AMTS)

Give the patient an address and ask them to repeat it at the end of the test.

1. How old are you?
2. What is the time to the nearest hour?
3. What year is it?
4. What is the name of this place?
5. Can the patient recognise two relevant people?
6. What is your date of birth?
7. Name the colours of the Australian flag or that of the patients own nationality.
8. Who is the present Prime Minister of Australia or the President of the USA or the King of England?
9. Count down from 20-1 (no errors, no clues).
10. Can you remember the address I gave you?

EACH QUESTION SCORES ONE POINT.

A SCORE OF LESS THAN 8 SUGGESTS ABNORMAL COGNITION

This may be delirium or dementia, inform the shift coordinator or medical officer.

Document the AMTS in the patient's medical record.

Adapted from: Hodgkinson H.M. Evaluation of a mental test for assessment of impairment in the elderly. Age Ageing 1972; 1:233-8. From Royal Perth Bentley Group Cognitive Impairment Working Party: October 2016